

## WHAT THIS IS

A desktop "command center" for assigning, observing, validating, and reviewing AI agent work across a workday -- a layer above individual agent sessions.

Long-term spec: `ref-docs/Generic-Agent-Management-Console-PRD.md`  
(drafted 2026-04-27)  
Active plan: `ref-docs/Project-Plan-2026-04-28.md`

## TIMELINE

- 2026-04-27 PRD drafted. Five exploratory views built in one evening (Cards, Agents, Graph, Board, Digest) on a Vite/React/Zustand prototype with mock data.
- 2026-04-28 Cards picked as the foundation. Three independent reviews commissioned (systems / software / agentic). Project Plan synthesized.
- 04-28 -> 29 Plan steps 1-5 landed in a single working session.
- 2026-04-29 Second review panel convened (same three reviewers plus a product designer) to plan Phases 6-11.

## ARCHITECTURE AS BUILT

- \* Frontend  
Vite + React 18 + TypeScript (strict) + Tailwind + Zustand.  
Three runtime deps total. Repository pattern in `src/data/repository.ts`; UI state split into `useUIStore`.
- \* Runtime backend  
`tools/console-server.mjs` -- standalone Node API on port 3001, owns the `.agent-console/` directory, spawns one `tools/claude-shim.mjs` daemon per active task.  
Vite proxies `/api/*` to it.
- \* Adapter  
Shim runs:  
`claude --print --output-format stream-json --verbose --resume <session_id>`  
per turn; daemon mode loops on a per-task message queue and writes events per `tools/protocol.md`.
- \* Data on disk  
`.agent-console/tasks/` + `runs/` + `agents/` -- JSON  
`messages/<id>.jsonl` -- queued user input  
`events.jsonl` -- append-only spine
- \* Tests  
51 (Vitest + jsdom + Testing Library). No Playwright yet -- deliberate.
- \* Deferred  
Tauri shell and SQLite event store (Phase 11) -- kept off until the protocol stabilizes against a second adapter.

## PLAN PROGRESS -- PHASES 1-5 DONE

- [done] Step 1 -- Type model  
Four explicit status axes: `lifecycleStatus` / `claimedStatus` / `validationStatus` / `reviewStatus`. Run separated from Task. `parentTaskId` added. Helpers: `taskBucket()`, `latestRun()`, `totalCost()`. Mock data migrated.
- [done] Step 2 -- Persistence boundary  
`src/data/repository.ts` + UI/data store split. Async API; in-memory backed at first; SQLite swap is now a one-file change.
- [done] Step 3 -- Componentization  
`CardsView` 499 -> 112 lines. `DetailPanel` 468 -> 87.  
New subdirs `cards/` and `detail/`. Every component <250 lines.

- [done] Step 4 -- Tests  
Vitest with 37 tests at the time (now 51) covering helpers and InMemoryRepository.
- [done] Step 5 -- Real Claude adapter end-to-end  
tools/claude-shim.mjs runs Claude Code via stream-json and writes .agent-console/ per tools/protocol.md. Vite plugin exposes /api/state and /api/messages; FileSystemRepository polls. Smoke-tested live: real Claude run, queued message drained on next invocation.

#### KEY PRODUCT DECISIONS MADE

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- \* Cards view is the foundation (decided 2026-04-28). The other four exploratory views are kept as alternate lenses; no significant work invested in them without confirmation.
- \* Three-axis status separation kept. Agent claim vs validator verdict vs human decision are explicitly distinct -- never collapsed.
- \* File-based protocol as the lingua franca -- .agent-console/ with task/run JSON + events.jsonl. Any runtime that can write a file can adapt in.
- \* Tauri deferred to Phase 11, after a second runtime adapter has stress-tested the protocol shape.
- \* SQLite event store deferred until file-protocol failure modes have shaped the schema (panel #2 trap-flag).

#### OPEN PUSHBACK ON THE PRD

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- \* Activity enum is too coarse -- needs richer types (tool\_call\_started, permission\_requested, thinking, subagent\_spawned) and a parentEventId. (-> Phase 7.2)
- \* auto/queue/interrupt is too clean -- interrupt\_after\_current\_tool is what runtimes actually deliver. Add cancel and pause; be honest in labels. (-> Phase 6.6)
- \* "Done today" framing is a trap -- digest should be a time window, not a day. (-> Phase 8.5)
- \* Deterministic-only validation won't survive research/docs. Compromise: deterministic gates 'verified'; LLM-as-judge gates 'judged\_ok'/'judged\_concerns'; never alone clears unverified. (-> Phase 9.3)
- \* Per-task token attribution is a comforting lie -- show run cost precisely, task cost as ~\$X sum-of-runs. (-> Phase 9.5)

#### WHAT'S NEXT -- PHASE 6 (ACTIVE)

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Goal: Close the dogfood loop. The single-shot shim is a demo, not a tool. Phase 6 makes the FS-mode UI usable as a daily Claude Code launcher.

- 6.1 Atomic writes (.tmp + rename) for shim writes -- partial-read window is empirically open today at 1Hz polling.
- 6.2 Fix messages/<id>.jsonl drain race via rename-on-drain.
- 6.3 Stream agentSummary + currentTool on each tool\_use so running cards aren't blank during the run.
- 6.4 Long-lived per-task shim daemon with claude --resume <session\_id>. The actual product unlock -- without resume, every "next turn" is amnesia.
- 6.5 Wire captureTask in FileSystemRepository so the UI can launch tasks. (Skip assign/accept/reject for now.)
- 6.6 Honest "Stop after current tool" (SIGINT) and "Cancel" (SIGTERM) buttons.
- 6.7 Extract API to tools/console-server.mjs standalone process; Vite proxies /api/\* to it.

## ROADMAP BEYOND PHASE 6

- Phase 7 -- Pay down structural debt while dogfooding  
Heartbeat sweep, richer activity parser with  
parentEventId, component tests, delete  
InMemoryRepository from runtime, wire mutation paths  
through queue.
- Phase 8 -- UX for daily use  
Live-activity rail, promote waitingOnUser with dock  
badge, progressive capture, decide on chat-card stack  
metaphor, fold Digest into Cards.
- Phase 9 -- Validation, contracts, cost  
Deterministic contract checkers, validation events,  
LLM-as-judge band, cost schema with cached-tokens /  
source flag.
- Phase 10 -- Second adapter, harden protocol  
codex / Responses / custom harness on the same  
protocol. seq INTEGER, idempotency, FSEvents  
reconciliation, sandbox for command\_exit.
- Phase 11 -- Tauri shell + SQLite  
Tauri v2 wrap, SQLite event store per PRD, native  
tray and notifications, replace 1Hz polling with  
fs-watch / SSE.

## EXPLICITLY NOT YET

- \* Graph view -- product designer's trap; most fun, least useful  
before live activity is solid.
- \* Workflow builder, multi-user review, issue tracker / PR  
integrations, LLM-providers cost-API, agent registry UI,  
suggested assignment, saved templates.
- \* Playwright -- defer until two adapters exist or Tauri ships.
- \* SQLite event store design -- build after the second adapter,  
not before.

## REVIEWER-FLAGGED TRAPS TO AVOID

- \* Don't harden the protocol broadly (seq / idempotency /  
lockfiles) before a week of real dogfooding. Phase 6.1 atomic  
writes is the deliberate exception.
- \* Don't port assignTask / acceptTask / rejectTask into  
FileSystemRepository before 6.1 + 6.7 land -- two writers  
without a coordinator will corrupt.
- \* Don't pretend mid-tool-call interrupt works. It doesn't.
- \* Don't add auth / CORS / multi-user / PID files to  
console-server.mjs -- bind to 127.0.0.1, single trusted user,  
until Tauri.

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End of summary  
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